

Point Forecasting Final Presentation

TDM Fall 2024 John Deere Parts Demand
Forecasting Class



The Data Mine



JOHN DEERE

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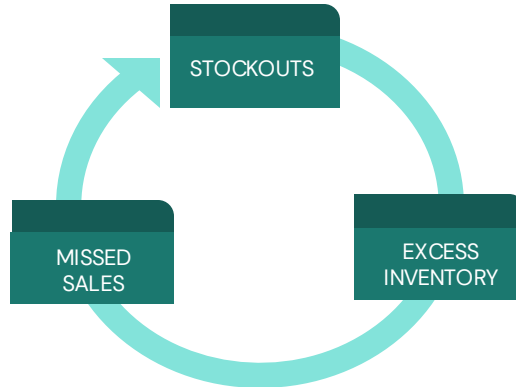
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**Limitations,
Future**

Business Problem

John Deere is facing significant challenges in accurately forecasting demand for its vast inventory of **1.6 million** part-location combinations. Due to the variability in demand patterns across different regions and product categories, traditional forecasting methods often fall short, leading to **stockouts, excess inventory, and missed sales opportunities**.

This lack of demand visibility not only impacts the company's ability to maintain optimal inventory levels but also affects **customer satisfaction and operational efficiency**.



Project Scope

Objective

Deliver a precise 12-month demand forecast for a part-location time series, minimizing overprediction and addressing any bias, to enhance inventory management



Business Problem

- ❑ Limited External Data Integration
- ❑ Ineffective Trend and Seasonality Capture
- ❑ Suboptimal Traditional Model Performance



Gap

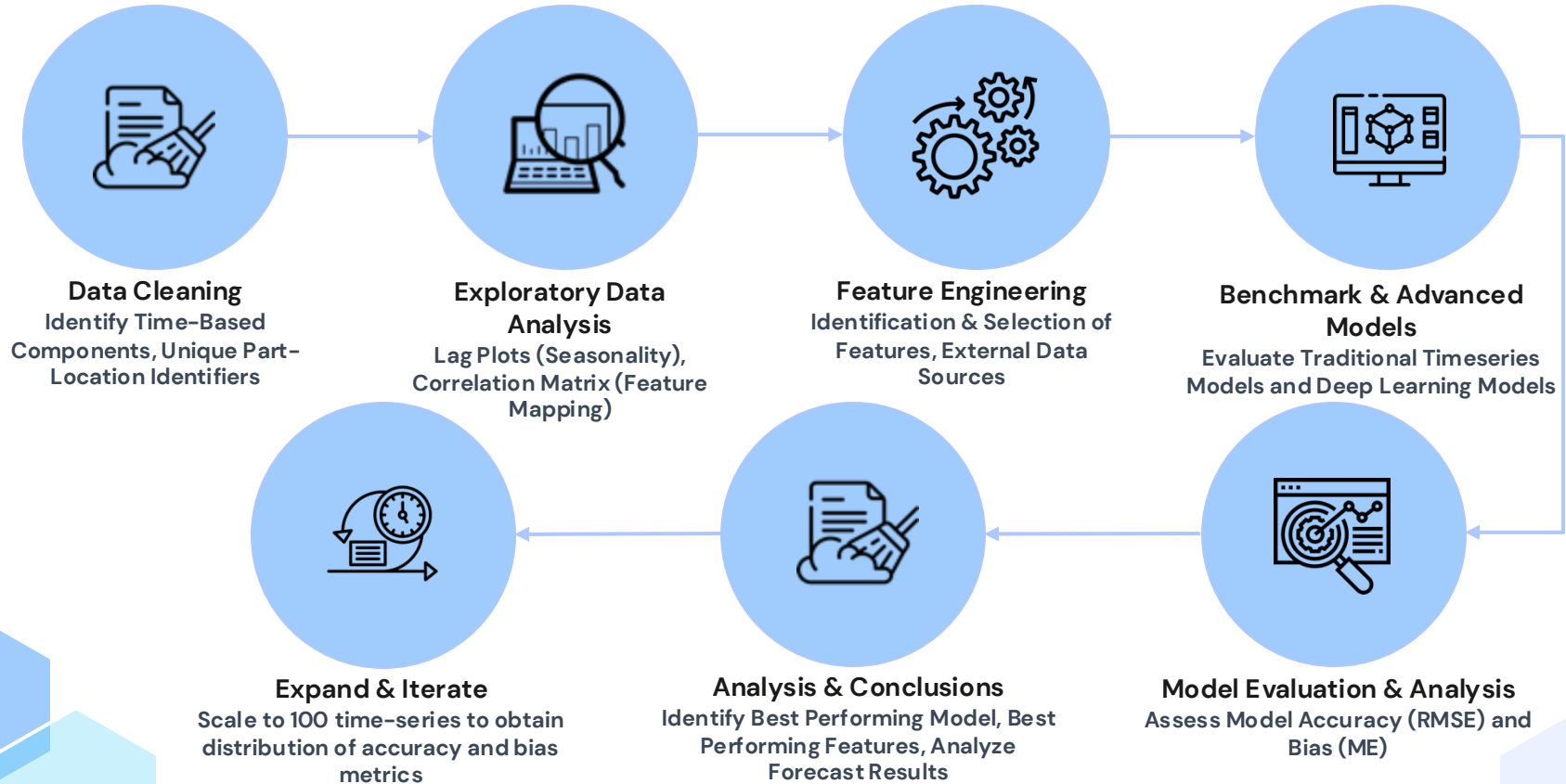
- ❑ Lack of External Factors (Interest Rate, Weather)
- ❑ Ineffective in capturing trend/seasonality effects
- ❑ Traditional vs. Machine Learning Models



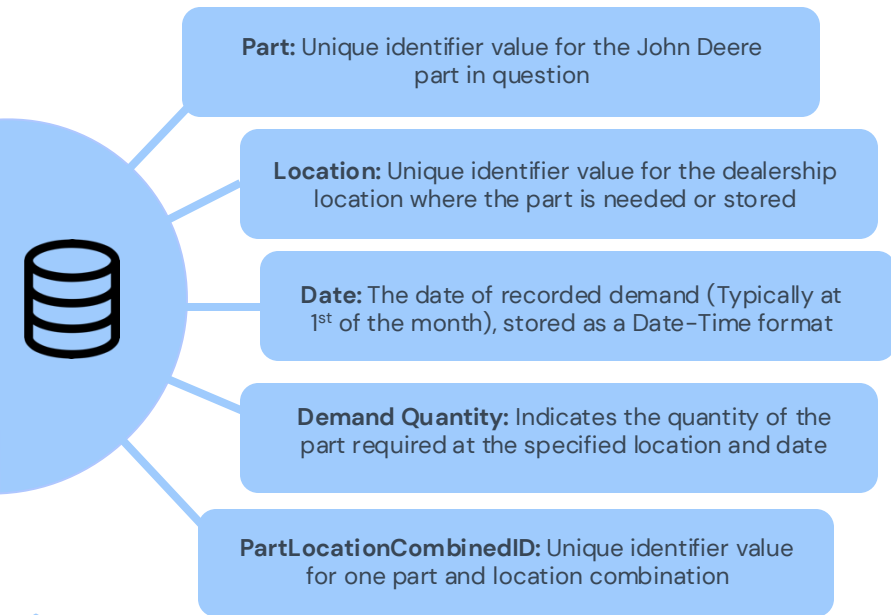
Deliverables & Value Proposition

- ❑ Integrate external market factors into the models
- ❑ Perform EDA, Trend & Seasonality Analysis
- ❑ Benchmark & compare with ML Models (Metrics)

Methodology



Historical (Internal) Dataset

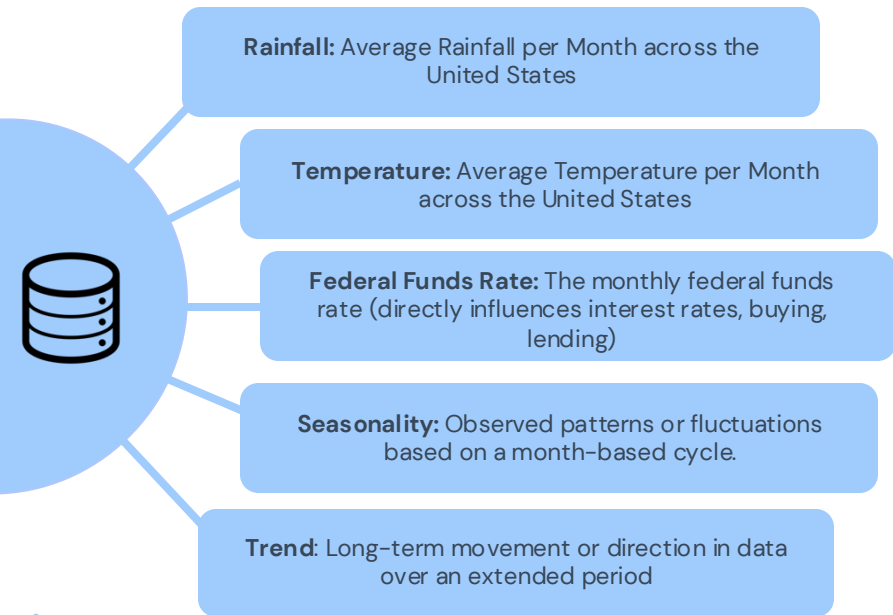


	Part	Location	Date	Quantity	PartLocationCombinationId	Year	Month_Num
0	1371	28803	2021-06-01	102	800	2021	6
1	1371	28803	2022-03-01	232	800	2022	3
2	1371	28803	2022-10-01	94	800	2022	10
3	1371	28803	2021-01-01	122	800	2021	1
4	1371	28803	2022-06-01	124	800	2022	6
5	1371	28803	2022-05-01	146	800	2022	5
6	1371	28803	2021-10-01	96	800	2021	10
7	1371	28803	2020-04-01	7	800	2020	4
8	1371	28803	2023-03-01	258	800	2023	3
9	1371	28803	2021-04-01	261	800	2021	4
10	1371	28803	2021-02-01	80	800	2021	2
11	1371	28803	2022-07-01	83	800	2022	7
12	1371	28803	2022-09-01	88	800	2022	9

Key Observations:

- ❑ The dataset analyzed focuses on a single part and location, as indicated by the constant values in the "Part" and "Location" columns.
- ❑ The "Quantity" column shows variability, with values ranging from 0 to 293, indicating fluctuations in demand over time (Normalization is important when considering more than 1 timeseries)

External Features



	Quantity	Year	Month	Seasonal	FEDFUNDS	Rain Value
0	5	2020	3	-32.192708	0.65	34.81
1	7	2020	4	-80.151042	0.05	34.12
2	0	2020	5	-1.317708	0.05	32.74
3	4	2020	6	86.640625	0.08	32.10
4	1	2020	7	-81.796875	0.09	32.28

Key Observations:

- ☐ The dataset uses Federal Funds Rate, Temperature, and Rainfall which are time-dependent variables
- ☐ For future predictions, online forecast models will be used as input to the demand forecasting models

Data Preprocessing

Why?

Data Cleaning & Quality

Remove missing, inconsistent, or outlier data to ensure accuracy and completeness.

Feature Engineering

Create and merge meaningful time-dependent features such as seasonality, trends, & external factors.

Identify Part-Location Combination

Group and find subset given unique part and location identifier.

Time Series Decomposition

Break down the time series data into trend, seasonality, month, year and residual components.

Layered Approach

Data Preprocessing Framework

One-Hot Encoding

Convert time-based categorical data to avoid month 12 vs month 1 bias

Merge External Factors

Combine Federal Funds, Avg Temperature, and Avg Rainfall Time Based Data

Train – Test – Split

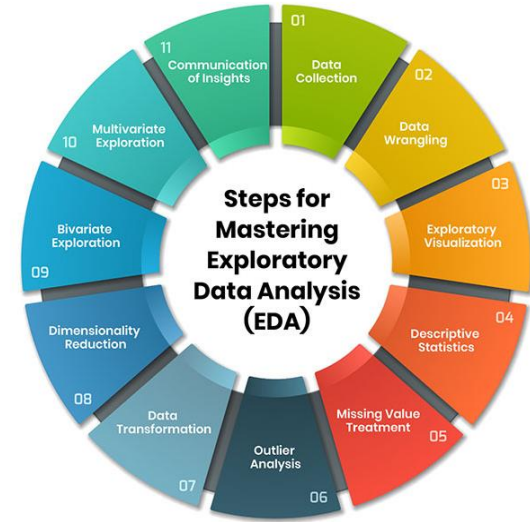
Use 80% of the dataset to train model, and test of 20% across models

Scaling & Normalization

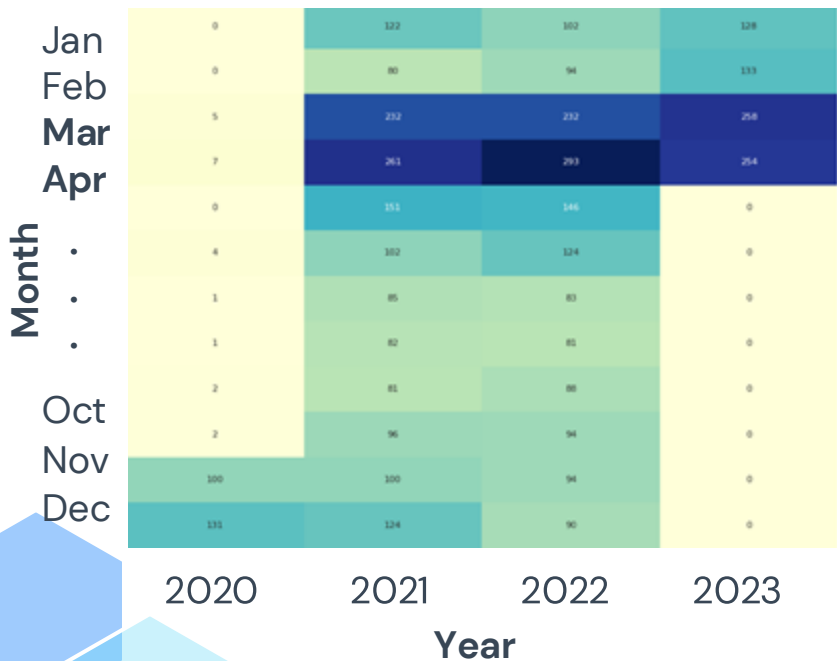
When extending to 100 timeseries, Qty Demanded affects RMSE value

Exploratory Data Analysis (EDA)

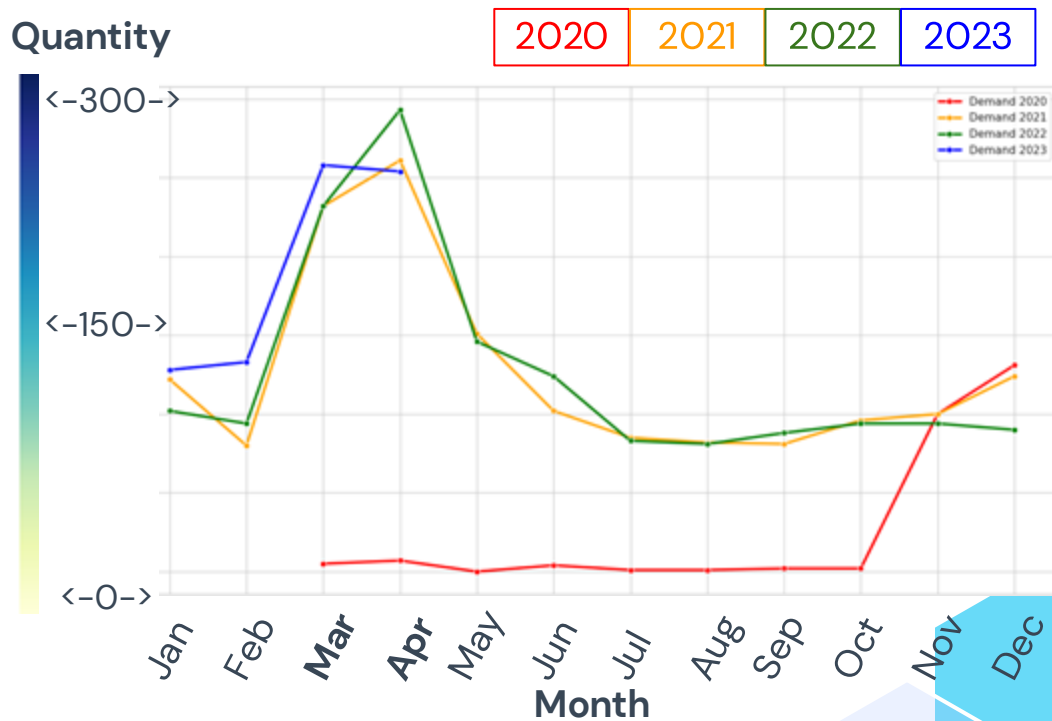
- ❑ **Trend and Anomaly Detection:** Identify and analyze current trends, patterns, and outliers in the data.
- ❑ **Correlation Analysis:** Examine relationships between key variables such as seasonality and trend influence the models, helping identify potential predictors for the model.
- ❑ **Seasonality and Variability:** Investigate recurring patterns (seasonality) and variability within the data, providing insight into predictable cycles and fluctuations.
- ❑ **Impact of External Variables:** Use correlation matrices and scatter plots to assess how external factors (e.g., weather, economic indicators) affect demand, highlighting their potential influence on model accuracy.



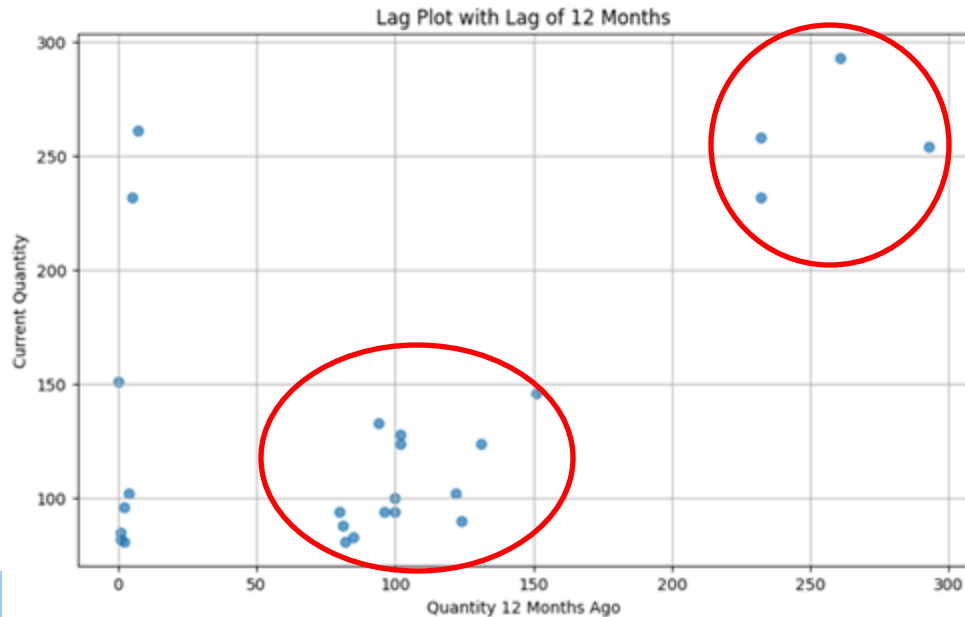
Substantial seasonal peaks in **April/March**



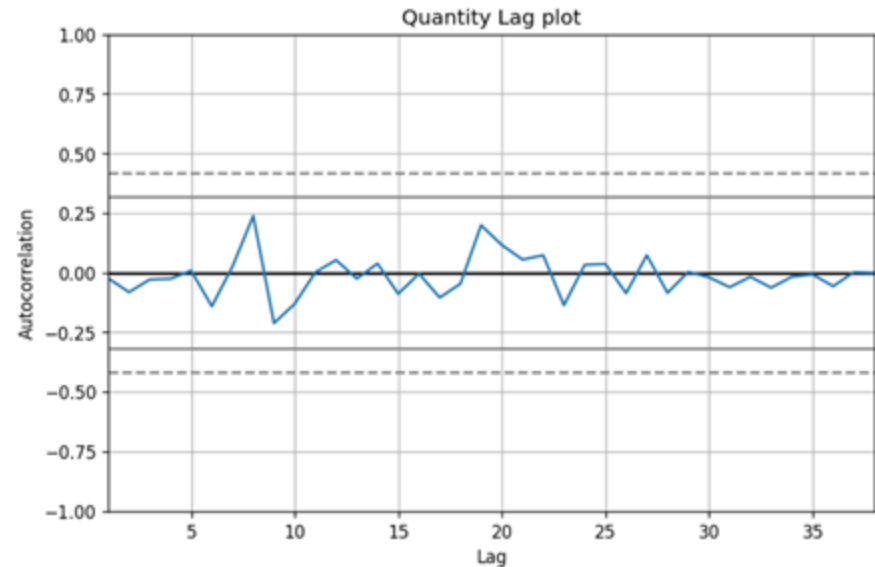
Quantity



Lag plots confirm this seasonality yet the data shows **weak autocorrelation**

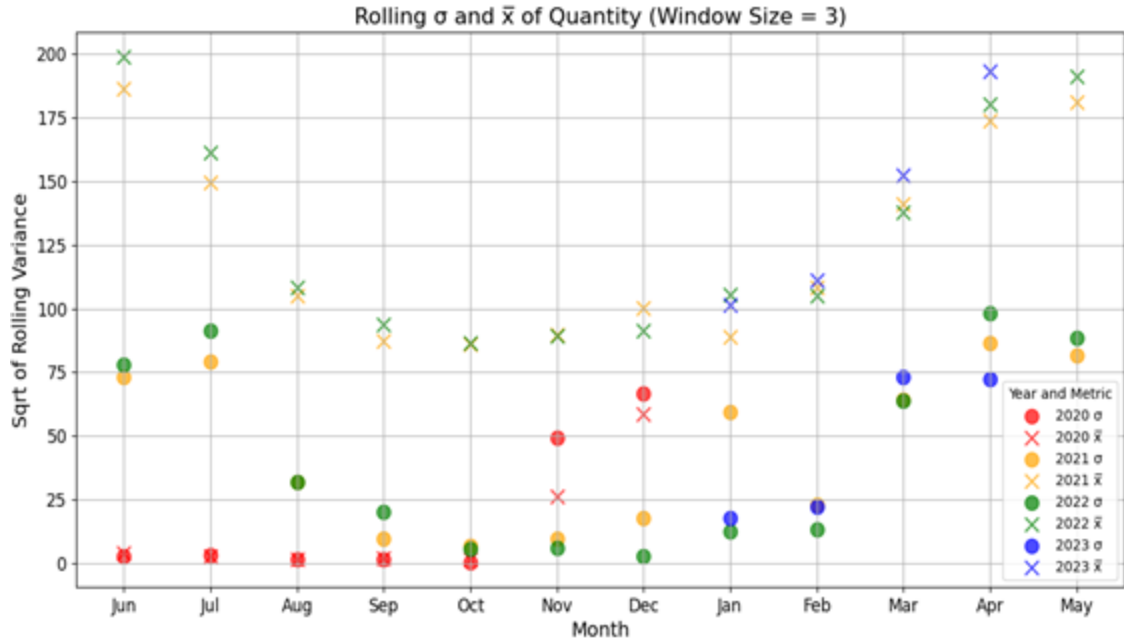
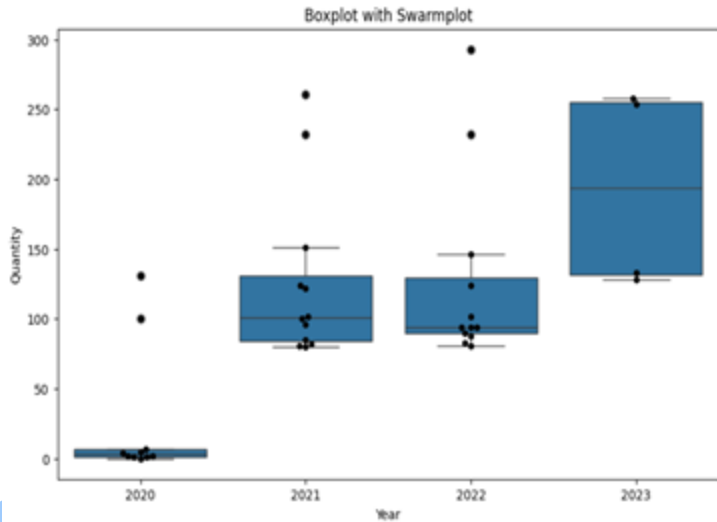


Clusters in the 12 month lag plot further demonstrate the seasonality in the dataset.

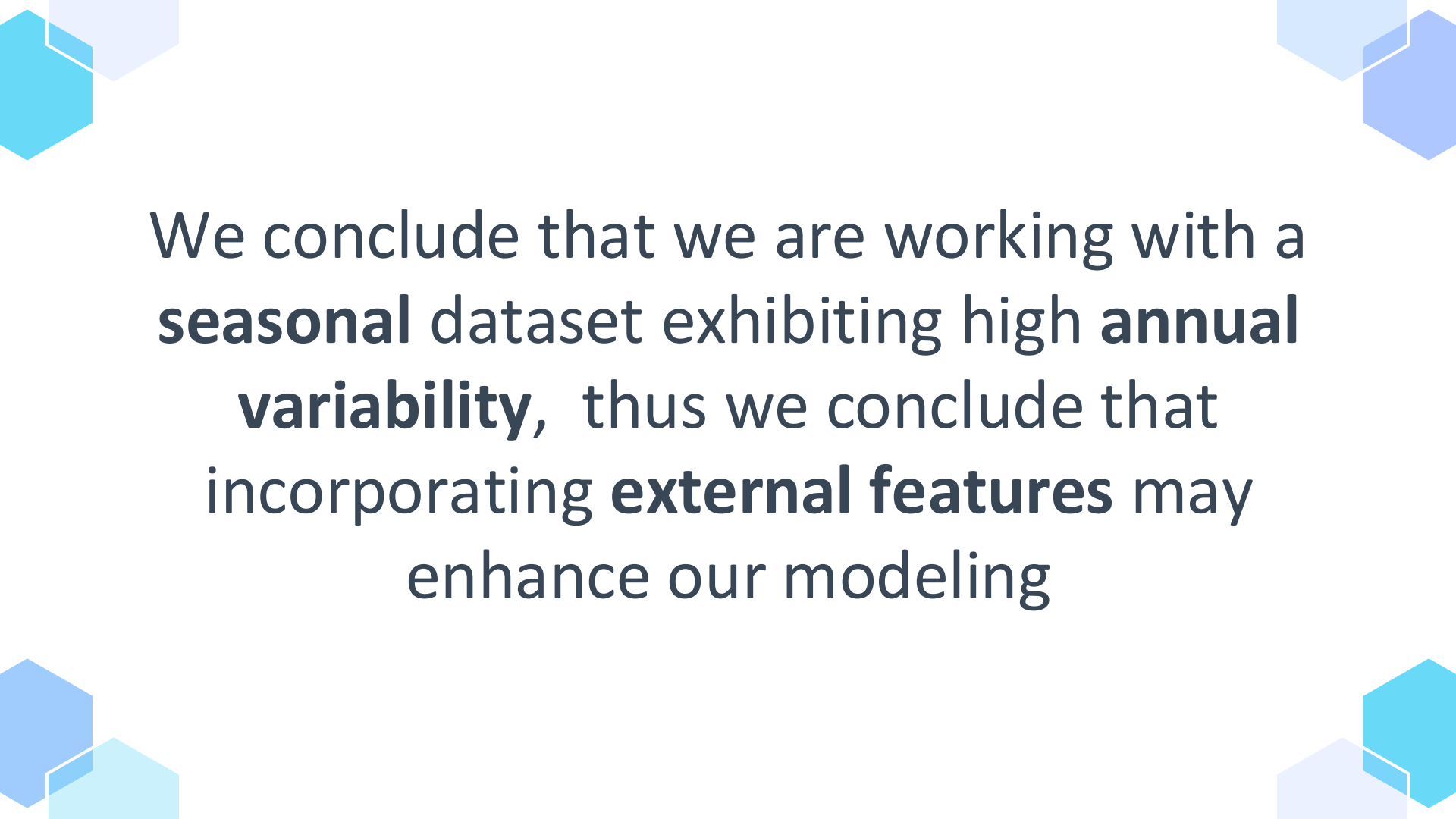


Minimal peaks indicate that observations do not highly rely on previous data

This has to do with the highly **interannual variability** within the data set

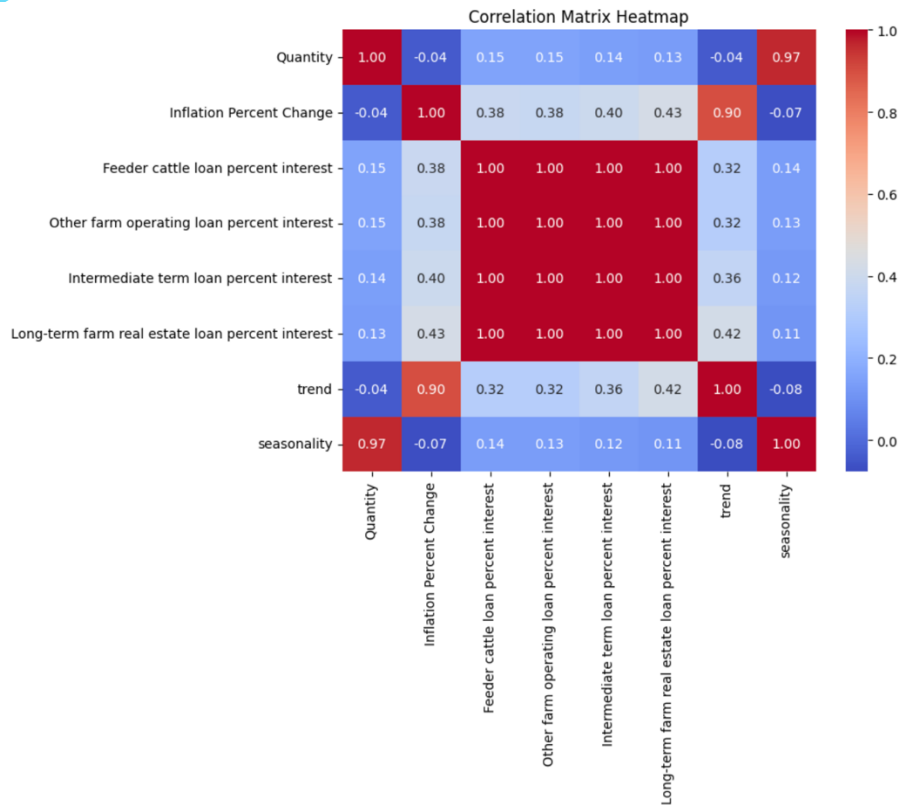


The data varies dramatically from year to year and month to month

The slide features a white background with decorative hexagonal shapes in the corners. The top-left corner has a light blue hexagon and a darker blue hexagon. The top-right corner has a light blue hexagon and a darker blue hexagon. The bottom-left corner has a light blue hexagon and a darker blue hexagon. The bottom-right corner has a light blue hexagon and a darker blue hexagon.

We conclude that we are working with a **seasonal** dataset exhibiting high **annual variability**, thus we conclude that incorporating **external features** may enhance our modeling

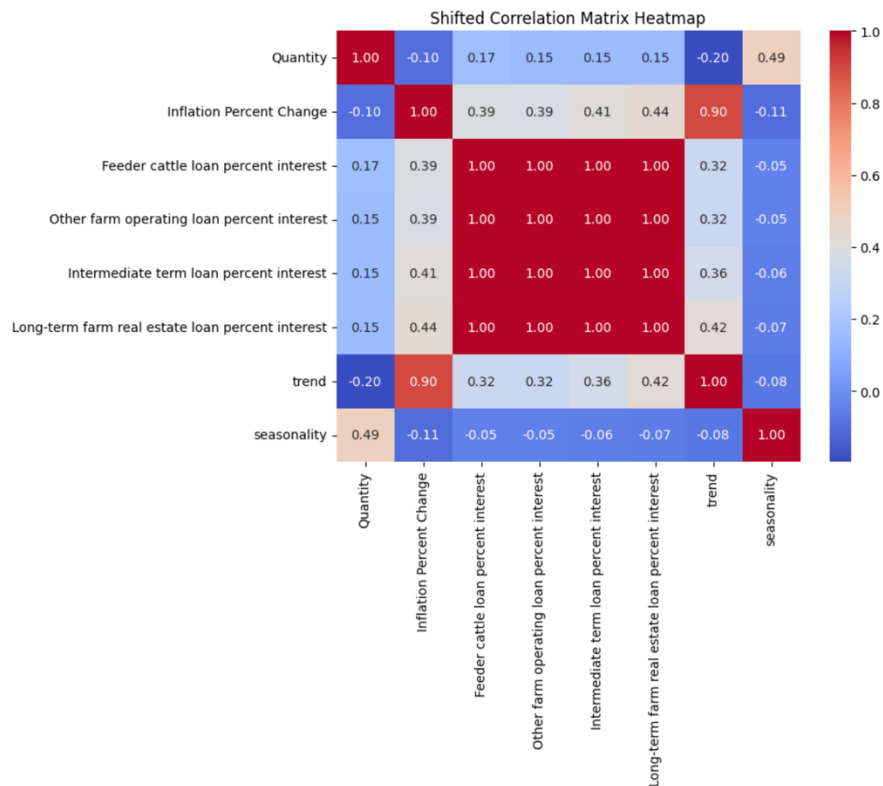
Linear Correlation Heatmap



Heatmap (Linear Correlation):

- Note the 1.00 Correlation across every Agriculture Loan Interest feature: This **external data was only seasonal**, so the **same exact data was copied over the three months in each season**, for every loan category
- Seasonality has the highest linear correlation with Demand Quantity of 0.97**
- Inflation Rate and Trend have linear correlation of 0.9. **We may be able to drop the supplemental Inflation Rate data**, as it overlaps with Trend

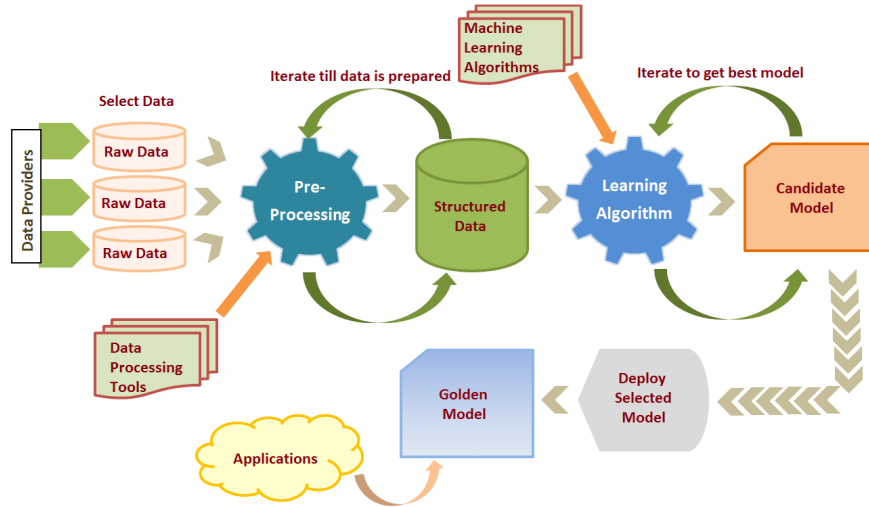
Lagged Correlation Heatmap



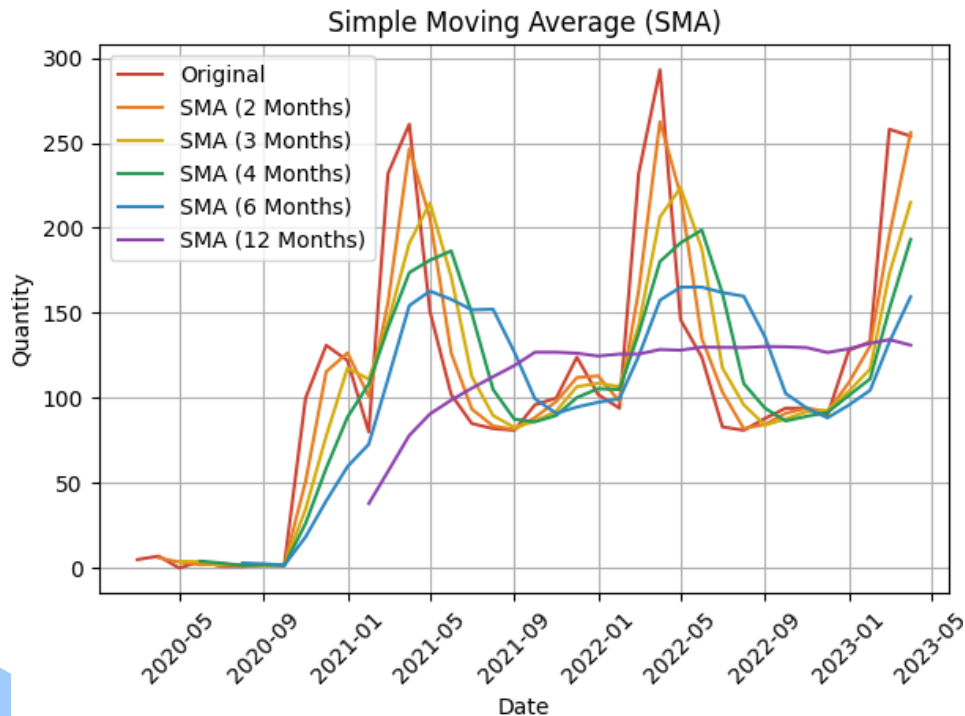
- Values are **lagged by 1-month**, i.e. current month's data will be connected to next month, to check if the current month's features could be predictors for the next month's demand.

- Linear Correlation between **Loan Interest Rates, Inflation Rate and Demand Quantity** goes up slightly compared to Unshifted (+0.01/0.02)
- Linear Correlation between **Seasonality and Demand Quantity** goes significantly down (-0.46)
- Moving forward, it may be wise to **lag the external features** (Loan Interest), while **keeping internal features** (Seasonality) **un-lagged**

Data Modelling

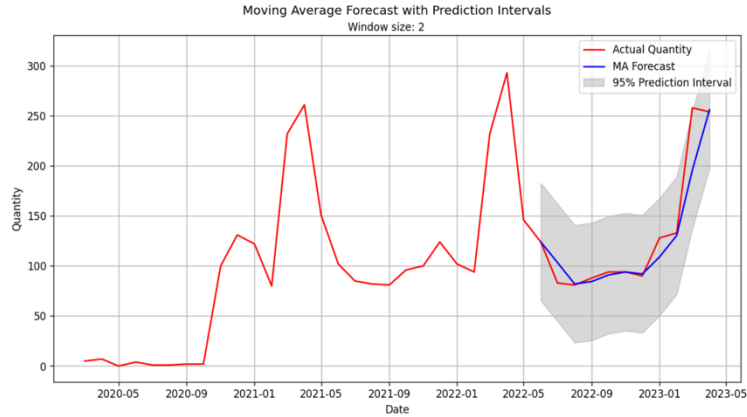


Moving Average

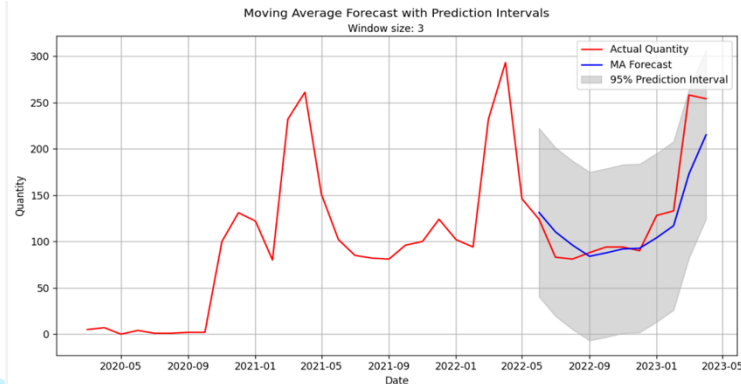


- Benchmark models based on simple techniques
- Baseline to compare complex models

Moving Average with Prediction Intervals

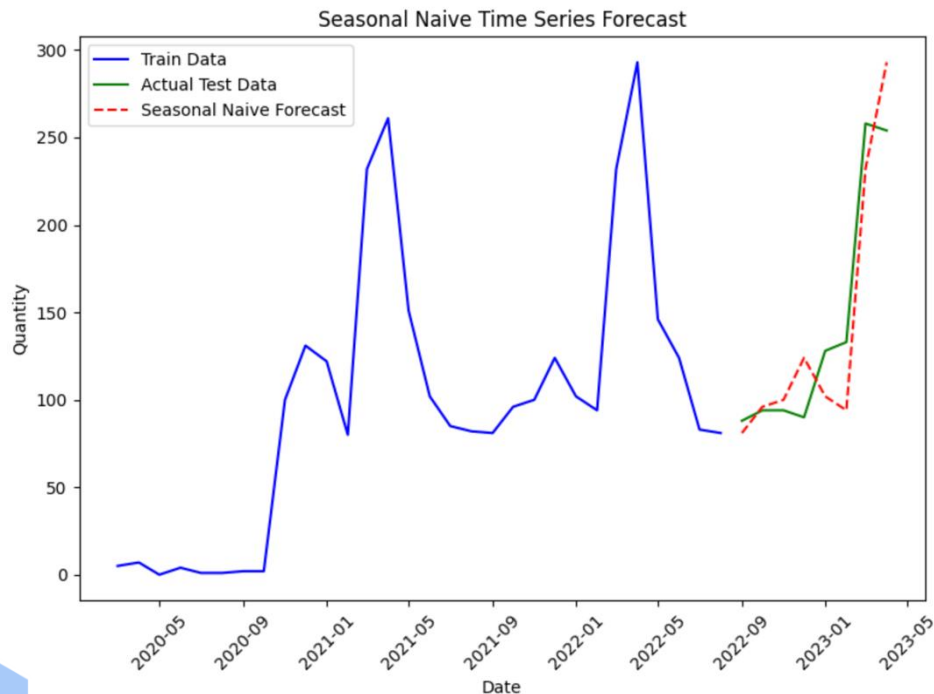


- Improved version of MA model (95% prediction interval)
- Prediction Interval – 95% chance forecasted values within range



- Assumption – Normality in data
- Window size – 2,3, & 4 months
- RMSE reduction by average 45%

Seasonal NAIVE

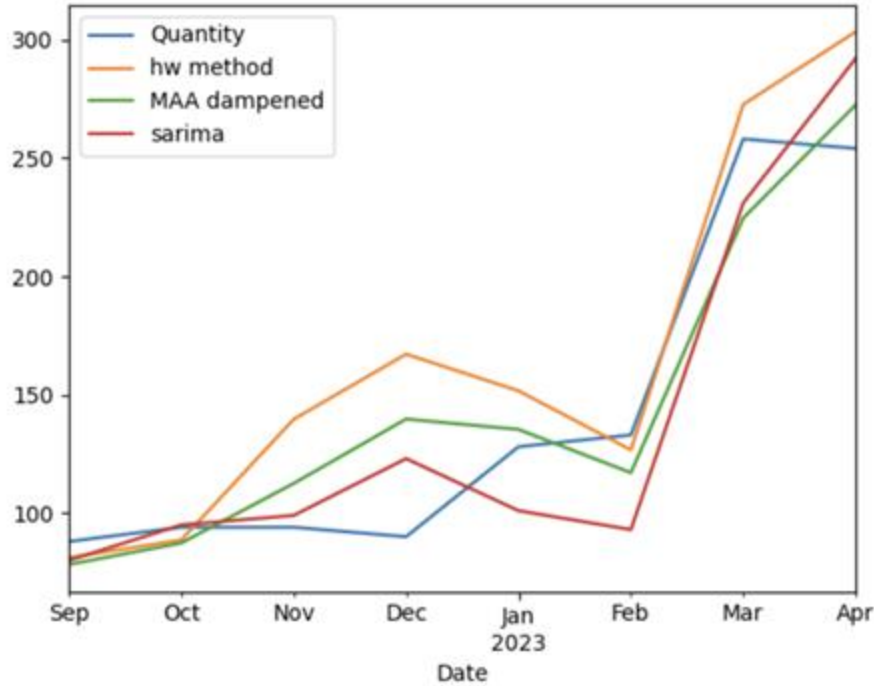


- Assumption – current year's demand resembles last year's demand pattern
- Fails to consider changing trends
- Benchmark model similar to MA

Error Metrics:

- **RMSE:** 26.55
- **ME:** 2.12
- **MAE:** 22.38
- **MSE:** 704.88

Exponential Smoothing



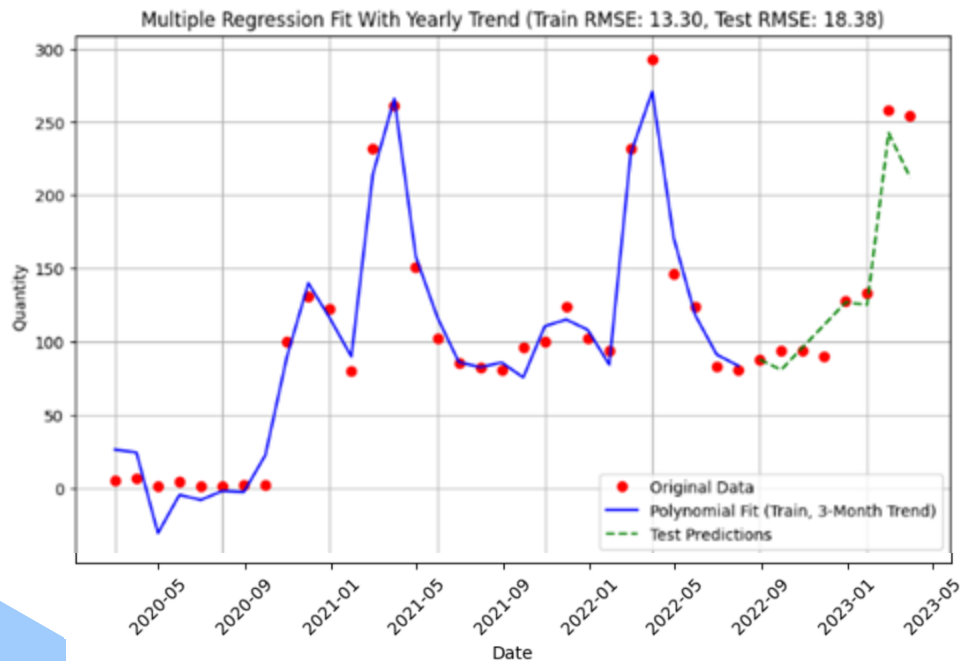
Model *	RMSE	ME
ANN	91.268	-61.375
AAN	79.643	-49.487
AAA	37.663 35	23.976
AADA	27.103	3.346
MADA	24.276	3.529

* ANN – level; AAN – level, trend; AAA – level, trend, seasonality, AADA – damped trend, MADA – multiplicative and damped trend

Exponential Smoothing

- **Observation** – AAA model captured seasonal peak for March
- **Overprediction** – Oct 2022 to Jan 2023
- **Negatively correlated** for Oct 2022 to Jan 2023
- Trend is captured accurately
- Seasonality correctly captured for Feb 2023 to Mar 2023
- Seasonality misses trough for Dec 2022

Regression

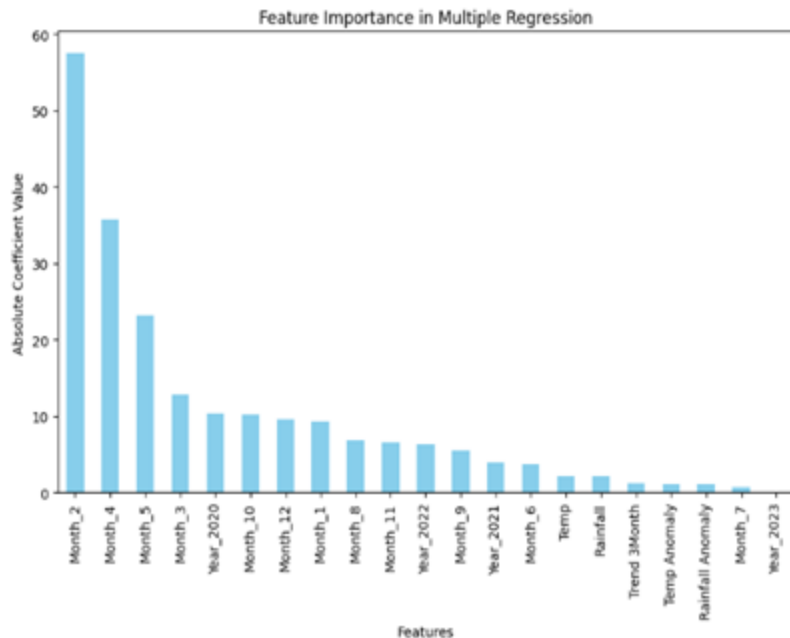


- Assignment of coefficients based on minimization of residual sum of squares

Error Metrics

- RMSE: 18.38
- ME : -7.05

Regression



- Rainfall significant feature

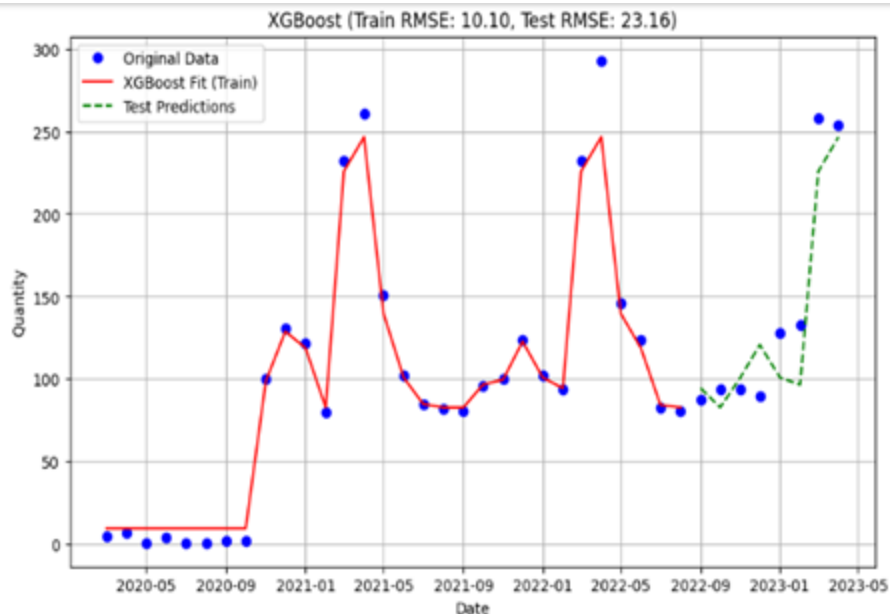
We believe that this is because customers could be more likely to use their equipment when there is less rain. The wear and tear from this use could cause them to order more parts.

XGBoost

Overview of XGBoost process



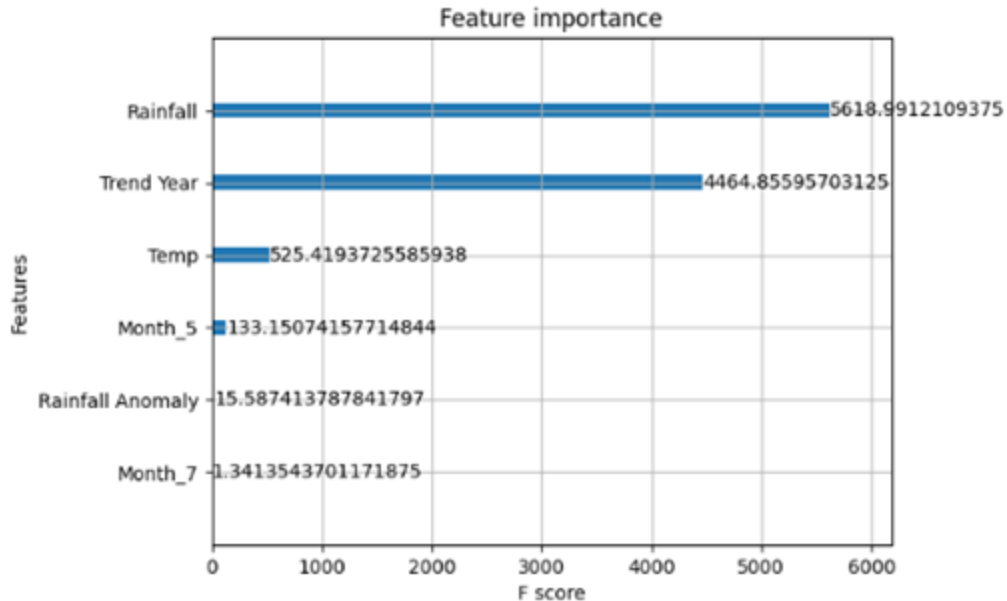
- Iteration of multiple weak learners
- Weak learners – decision trees



Error Metrics

- RMSE: 23.16
- ME: -8.65

XGBoost

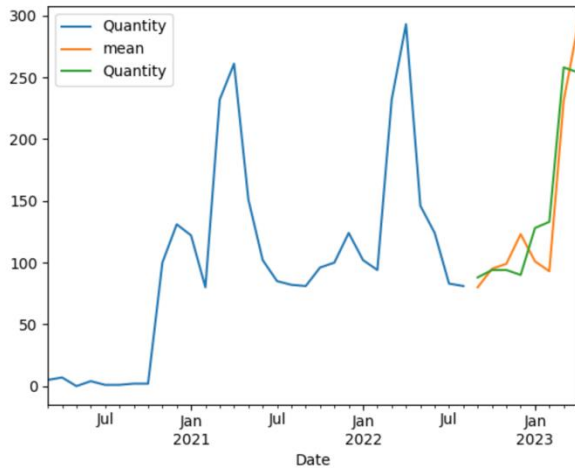


- Rainfall, yearly trend, and temperature are important features

- Consist with feature importance chart of regression model

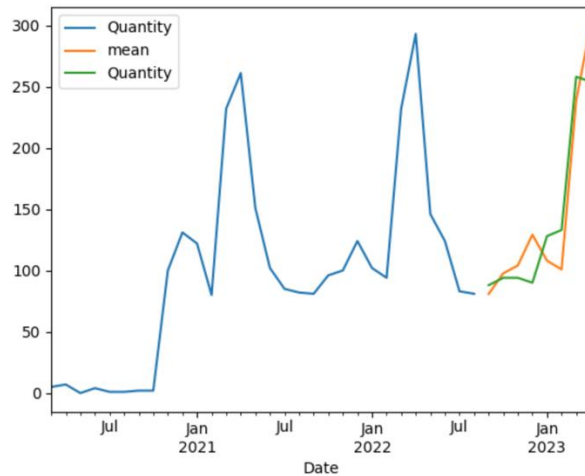
SARIMA/SARIMAX

SARIMA Forecast (No External Features)



SARIMA(0,1,0) (0,1,0,12)

SARIMA Forecast (Federal Funds External Features)



SARIMA(0,1,0) (0,1,0,12) + Federal Funds

- Addition of Federal Funds as a feature improved model accuracy and helped eliminate underprediction

Model	RMSE	ME
RTF	35.827	-15.756
RF	28.623	-5.045
R	28.744	-5.722
SARIMA	26.648	-3.125
F	26.490	2.514

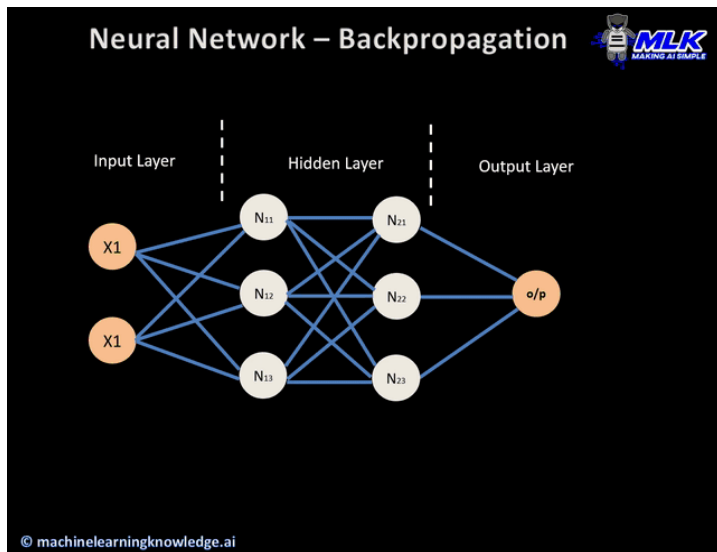
- RTF – Rainfall, Temperature, Federal Funds
- RF – Rainfall, Temperature
- R – Rainfall
- SARIMA – No External Features Used
- F – Federal Funds

SARIMA/ SARIMAX

- External predictors into the SARIMA model, like Rainfall and Temperature, **increased** the RMSE of the model and produced a **negative** ME. Similar effect was observed with the introduction of Rainfall Anomaly and Temperature anomaly
- The only predictor that had a positive impact on the performance of the model was the inclusion of the federal funds rate. Improving the RMSE and producing a ME that was indicative of more overprediction from the model

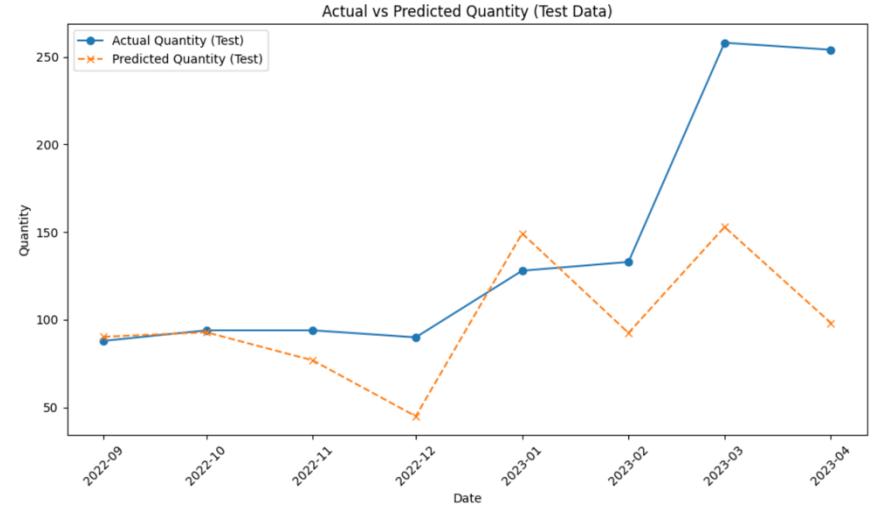
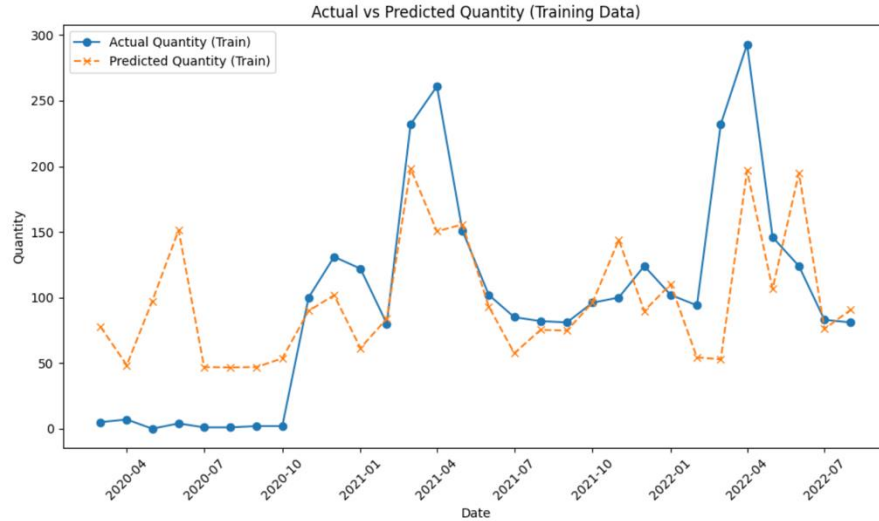
Model	RMSE	ME
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How Does a Neural Network Work?

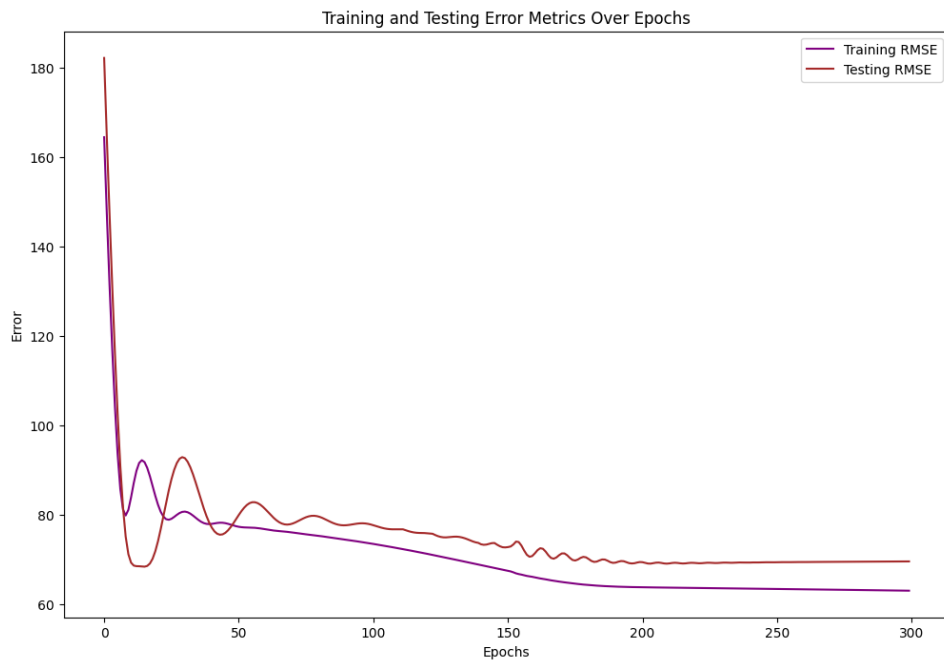


- **Input Layer** (Features – Rainfall, temp.), **Hidden Layer** (Analysis & Processing), **Output Layer** (Predicted Demand Quantity)
 - Weighted Sum of Inputs
 - **Activation Function:** introduces non-linearity, allowing the network to model complex patterns and make decisions
-
- **Hyperparameters:** Activation Function: ReLu, Learning Rate: 0.001, 300 Epochs
 - **Challenge:** Performance impact due to limited data

Neural Networks Results



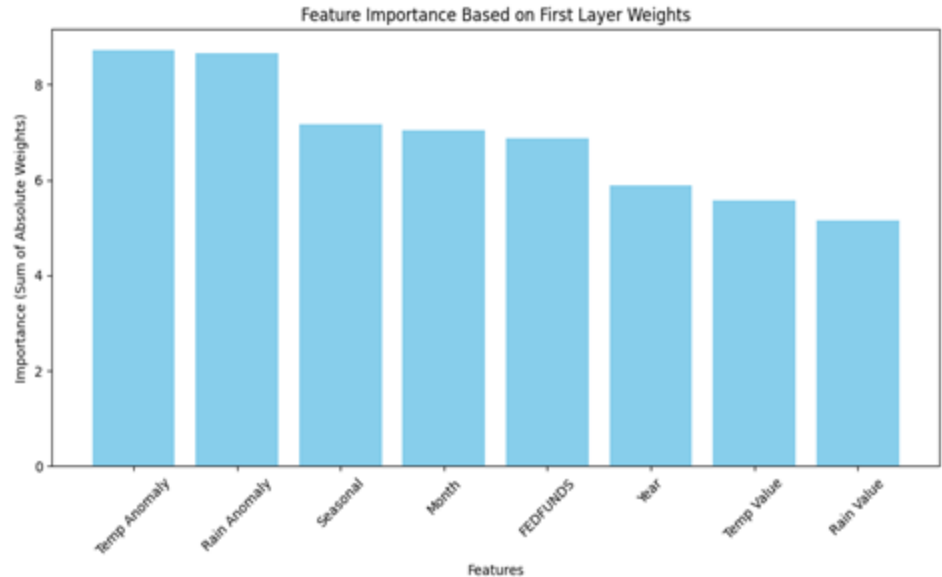
Neural Network Error



Training MSE	3965.2439
Training MAE	45.9136
Training RMSE	62.9702
Testing MSE	4834.1484
Testing MAE	47.7635
Testing RMSE	69.5280

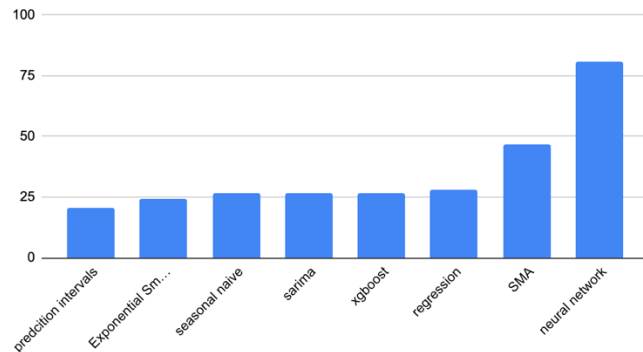
Neural Networks Feature Importance

- **Key Features Identified:** Seasonality, federal funds rate, temperature, and rainfall
- **Model Limitations:** Limited data and poor predictions impacted accuracy of feature importance rankings
- **Caution in Interpretation:** Feature importance results might not fully reflect true impact due to data constraints

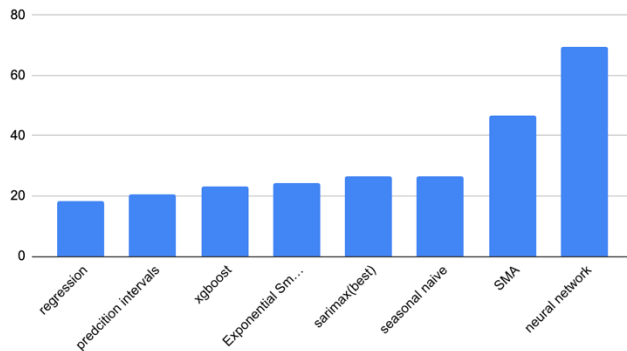


Evaluation

RMSE(internal features only)



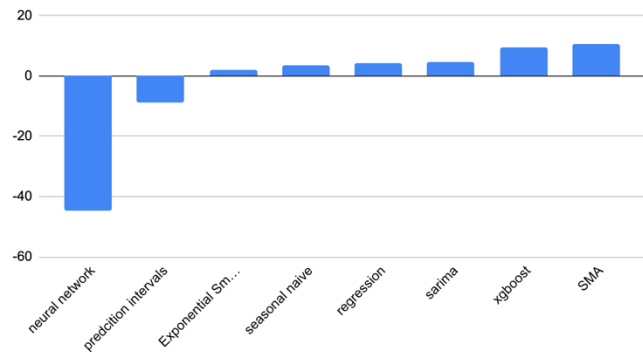
RMSE(Using External Features)



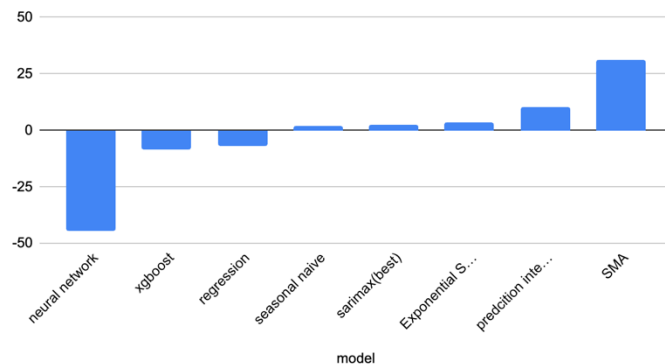
- ❑ Almost all models **outperformed** benchmark simple moving average model
- ❑ Exponential Smoothing and XGBoost showed **strong performance against benchmark model** and serve as strong foundation for further model development
- ❑ Testing on additional time series can further validate model results

Evaluation

ME(Only internal features)



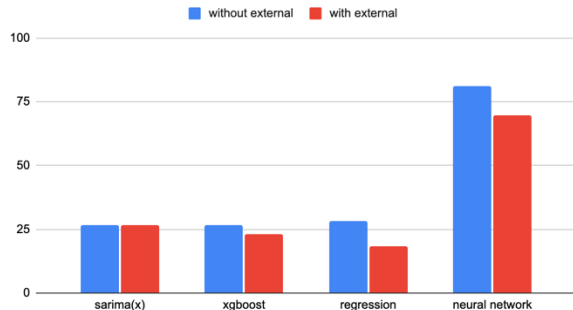
ME(Using External Features)



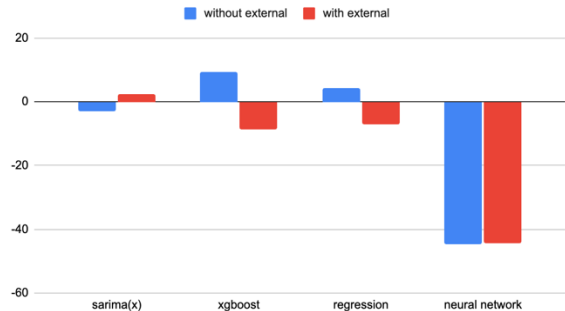
- ❑ Almost all models without external features showed consistent **overprediction** as compared to benchmark models
- ❑ Systemic underprediction can cause locations to **undersupply** and fail to capture real consumer demand. Results in lost revenue and produces poor data for future use.
- ❑ Consistent **overprediction** shows the need of further model development.

External Features Impact on Model Performance

RMSE(+/- external features)



ME(+/- External Features)



- ❑ **SARIMA, XGBoost, Regression, & Neural Networks** which were the models that could **incorporate external features** clearly showed a **decrease in RMSE**
- ❑ On the other hand there was also a **greater propensity for underprediction** following such improvements in RMSE reflected in the ME
- ❑ This could suggest that there does need to be further development of external features, as while RMSE was increased, underprediction can be more costly than less correct overpredictions

Results and Conclusion

- Multiple regression (MLR) – least RMSE with a value of **18.63**.
- Exponential smoothing model had the best ME with a value of **3.529**
- Neural Network : the highest ME value at **~50** and highest RMSE value at **69.5280**. This is due to limited data for training
- **Conclusion:** Although MLR performed the best, generalizing would require testing on multiple part location combinations



Limitations and Future Scope

- **Limitations**

- Trained only on a single part location combination and only three years worth of data
- Feature are time dependent, should have tested by lagging certain features for prediction

- **Future Scope**

- Extend it to 100 part – location combination
- Probabilistic modelling – provides a range of expected demand values
 - Quantile Regression, Bayesian Structural Time Series